

Segmented-Shroud Aeromechanics

A guide to the whole project: the question, the physics, the method, the code, and every mistake made so far

Repository **500ft/segmented-shroud-aeromechanics** · state as of 27 September 2026 · 96 commits · 216 tests

Read this first. This guide teaches the project as it actually is, not as a tidy summary. Where the repository got something wrong, the error is here with the reasoning that produced it and the counterexample that killed it. That is deliberate. The errors are the most transferable part of the project, because every one of them is a mistake that looks like competence from the inside. Nothing in this repository is a measurement. No rotor has been built, no rig has run, and the one computational case that finished has an unresolved comparison against its experiment.

How to use it

Parts I and II are the foundations: what is being asked and how computational claims are made credible. Part III is the experiment design and its statistics. Part IV is the evidence discipline, which is the part most people skip and the part that does the most work here. Part V walks the code. Part VI is the history and the error catalogue. Part VII is a study path with exercises against the live repository.

You do not need to read it in order, but Part II before Part VI will make the error catalogue land properly. Terms are defined the first time they appear.

Contents

Part	Subject	What you will be able to do afterwards
I	Orientation	State the research question precisely and say why each clause is in it
II	Credibility of computation	Distinguish verification from validation and compute a grid-convergence uncertainty
III	The experiment	Read a design matrix, spot an unidentifiable model, and classify a result three ways
IV	Evidence discipline	Trace any number to its provenance and know what may be said out loud
V	The code	Navigate all seventeen scripts and understand the test philosophy
VI	History and errors	Recognise the failure modes before you repeat them
VII	Study path	Work through the repository yourself with exercises and sources

PART I

Orientation: what is actually being asked

Before any solver or statistic, you need the question in a form that could be answered wrongly. Most of the work in the first three weeks of this project went into making the question falsifiable.

1. The question, and why it is worded that way

The owner's original wording was:

At equal mean tip clearance, do discrete circumferential seams in a ducted-rotor shroud change thrust, efficiency, or tonal noise relative to a continuous duct, and is the effect predictable from seam count, gap, and misalignment?

That is a good engineering question and an unanswerable research question, for two reasons that are worth understanding because they recur everywhere.

1.1 "Equal mean clearance" has no meaning until you say how you average

Tip clearance is the radial gap between the blade tip and the shroud wall. Over a seam there is *no wall*, so there is no clearance there at all. Not zero, not large: undefined. Any statement of the form "both configurations have the same mean clearance" is therefore incomplete until you declare what the mean is taken over.

The repository admits three treatments: a wall-only mean that integrates over the walled angular domain and ignores seams; a fixed angular grid with declared occlusion, where seam samples are recorded as missing rather than dropped; or both, reported side by side. Decision D5 requires every computational case to be post-processed under both rules and the agreement reported. Simulation can do this cheaply on identical geometry, which no physical rig can, and that is one of the study's jobs rather than a footnote.

Why this matters far beyond this project. Whenever a control condition is defined by "holding X equal", ask what happens to X where it does not exist. A comparison that silently changes its own definition of the matched quantity is not a controlled comparison, and the defect is invisible in the results.

1.2 Noise had to leave the headline

Tonal noise is the sound at blade-passing frequency and its harmonics. It is genuinely interesting for a shrouded rotor and it cannot be produced by the method this project can afford. Steady Reynolds-averaged simulation solves for a time-averaged flow; a tone is an unsteady pressure fluctuation. Asking a steady method for a tone is a category error, not a resolution problem. Noise is therefore a secondary observable of the final study and a trigger-gated extension, not part of the primary question.

1.3 The revised question

At equal mean tip clearance under a declared averaging rule, and at matched thrust, do discrete circumferential seams (count n , gap g , radial step s) change shaft power and thrust relative to a continuous duct by MORE THAN the combined numerical and measurement uncertainty, and does a descriptor model in (n, g, s) predict a HELD-OUT seam configuration better than a mean-clearance model?

Three additions carry all the weight. **Matched thrust** fixes what is held constant while power is compared, because comparing power at equal rotational speed compares two machines doing different amounts of work. **More than the combined uncertainty** makes the claim falsifiable and forbids reporting a difference smaller than the error bars as a finding. **Held-out** makes the second clause a genuine prediction rather than a curve fit: the model is frozen, then shown a configuration it has never seen.

2. The physics you need

2.1 Why a duct at all

A ducted rotor is a rotor inside a shroud. For the same diameter and power in hover it can produce more thrust than an open propeller, because the duct's inlet lip carries suction that adds thrust, and the duct suppresses the tip vortex that otherwise rolls up off the blade tip and wastes energy. Small vertical-lift aircraft use them for this reason, and also because a shroud is a guard around a spinning blade.

2.2 Why the tip gap is the interesting part

The blade tip must not touch the duct, so there is always a gap. Across that gap sits a pressure difference between the blade's pressure and suction sides, which drives a leakage jet over the tip. That jet rolls into a tip-leakage vortex which is one of the dominant loss mechanisms in the machine. Smaller gap, less leakage, better performance. The whole design tension is that you want the gap as small as manufacturing, thermal growth, vibration and landing loads will allow.

2.3 Why seams are the project's actual subject

A shroud that folds, deploys or is assembled from segments has *joints*. A joint can be a through-wall seam, an open slot in the wall, or a blind groove that does not pierce it. Each creates a local departure from a smooth circular wall: a missing wall over some angle, and possibly a radial step where two segments do not line up.

The hypothesis is that the *arrangement* of that missing material matters, not merely the average gap it produces. If true, a designer can choose where to put joints. If false, the design rule collapses to "keep the mean clearance small" and the arrangement is free. Both outcomes are useful; only one is interesting.

A distinction the repository had to learn. A through-wall seam and a blind casing groove are not the same object. Compressor literature on casing treatment uses grooves that do not pierce the wall. Treating that literature as direct evidence for open seams was corrected on 25 September: it is an analogy, not an equivalence, and the design contract now distinguishes the two explicitly.

3. Why this is hard

The effect being chased is small. The clearance change a seam introduces is a fraction of a millimetre on a gap that is itself around a millimetre. Turn that into a power difference between two configurations and you are looking for a difference of a few percent, measured on a machine whose own run-to-run variation may

be of the same order.

That produces three separate problems, and the project keeps them separate on purpose:

- 1 **Can the simulation be trusted?** A solver will always return a number. Whether that number corresponds to reality is a separate question with its own procedure. This is Part II.
- 2 **Can the experiment resolve the effect?** This is a measurement-uncertainty question, answered by a budget of every contributing term, and the budget is currently incomplete.
- 3 **Is the effect worth knowing?** A difference can be real, measurable and too small to change any design decision. That threshold is an owner decision and it has not been made.

Conflating these is the standard failure. A result reported as "validated" usually means the first question was answered loosely and the third was never asked.

PART II

Making a computation credible

This is the part with a procedure. Everything here is standard practice in computational fluid dynamics, and the project follows it deliberately rather than inventing project-specific rules.

4. Verification and validation are different questions

This distinction does more work than any other idea in the project, and almost every error in Part VI is a failure to hold it.

	Verification	Validation
Question	Am I solving the equations correctly?	Am I solving the correct equations?
Compares against	Itself, on finer and finer grids	Physical measurement
Failure means	A coding, meshing or convergence problem	The model does not describe the world
Can be done with	No experiment at all	Only with experimental data <i>and its uncertainty</i>

A computation can be perfectly verified and completely wrong. It can also agree with an experiment for compensating reasons. The procedures are separate, run in order, and reported separately.

The word "validated" is load-bearing. In this repository nothing may be described as validated without naming the quantity, the conditions and the band. On 25 September the phrase "A0.1 validates a two-dimensional airfoil workflow" was removed from both the report and the code, because the case's own verdict was that its uncertainty budget was incomplete. A claim of validation while a required uncertainty term is unquantified is not a weak claim; it is an unsupported one.

5. The validation ladder

You do not validate a ducted rotor with seams by simulating a ducted rotor with seams and declaring success. You climb from cases where public measurements exist:

Rung	Case	What it can establish	State
A0.1	Two-dimensional NACA 0012 aerofoil, NASA Turbulence Modeling Resource conditions	Mesh conversion, boundary conditions, freestream turbulence and solver settings on a case with public data	Run. Comparison against the experiment unresolved
A0.2	Caradonna–Tung hovering rotor	Rotating-frame loading in three dimensions	Not started; solver regime decision open
A0.3	A ducted fan	Tip-gap flow inside a duct	Deferred: no source with usable geometry confirmed

Every result from the main study carries the label *rotor-validated*, *gap-unvalidated* until the third rung exists. That label is not modesty. It states precisely which physics has been checked against reality and which has not.

The baseline rotor was chosen by an explicit owner rule: take the most-used case, so the most published data exists to compare against. That trades a narrower validation claim for a much richer comparison set, and the trade is recorded rather than assumed.

6. Grid convergence, worked with the project's real numbers

A discretised solution carries error from the mesh being finite. Richardson extrapolation and the grid-convergence index estimate how much. The procedure needs at least three systematically refined grids at a refinement ratio of at least 1.3.

6.1 The three grids

Grid	Cells	Lift coefficient
fine (g3)	57,344	1.098833
medium (g2)	14,336	1.091606
coarse (g1)	3,584	1.045623

Cell count quadruples each step, so in two dimensions the representative cell size halves and the refinement ratio is exactly 2.

6.2 The apparent order of convergence

The formal order is what the numerical scheme promises, here about second order. The *apparent* order is what the three solutions actually exhibit, solved by fixed-point iteration from the two differences. Computing it rather than assuming it is the whole point: assuming second order when the solution is behaving differently hides the error you are trying to estimate.

```
e21 = phi2 - phi1      e32 = phi3 - phi2
s  = sign(e32 / e21)   # +1 monotonic, -1 oscillatory
p  = | ln|e32/e21| + q(p) | / ln(r21)      solved iteratively

A0.1 result:  p = 2.6697,  monotonic
```

A sign change between the two differences means the solution is oscillating between grid levels rather than converging, and the estimate is not usable. In this project that case produces the verdict **INCONCLUSIVE**, and it outranks everything else, because if the numerical evidence is unusable there is nothing to compare.

6.3 The index

```
e_a21 = |(phi1 - phi2) / phi1|          relative difference
GCI21 = Fs * e_a21 / (r21^p - 1)        Fs = 1.25

      = 1.25 * 0.006577 / (2^2.6697 - 1)
      = 0.153 %   of the fine-grid lift
```

Read the index as an uncertainty band on the fine-grid answer arising from the mesh alone, roughly at a 95 percent level. Around one part in six hundred is a well-converged two-dimensional case.

The safety factor is used outside its stated condition. Celik and colleagues prescribe 1.25 where the apparent order is close to the formal order. Here it is 2.6697 against a formal order near 2, a third above it, and the code does not limit it. The project computed the consequence rather than arguing about it: across defensible combinations of factor and order limiting, the index runs from 0.153 to 0.658 percent. The comparison error is 2.22 percent, three to fifteen times larger, so the choice changes nothing here. That is recorded beside the result, and a test recomputes the table from the committed record so it cannot rot.

7. Uncertainty, and the rule that a missing term is not zero

The comparison error is the simulation minus the data. Whether that difference means anything depends on how well you know both sides. The standard combination is:

```
E      = S - D                                simulation minus data
U_val = sqrt( U_num^2 + U_input^2 + U_D^2 )

U_num   numerical, from the grid study
U_input  from inputs that are assumed or derived rather than verified
U_D      experimental uncertainty, as reported by the source
```

Quadrature is justified only because these are estimates of independent error sources. That condition matters, and violating it is one of the corrected errors in Part VI.

7.1 What A0.1 actually produced

Quantity	Value	Comment
Simulation, fine grid	1.098833	
Experiment	1.07502	NASA curated tripped data at the measured angle 10.12°
Comparison error E	+0.023813	+2.22 % of the experimental value
U_num	0.001684	from the grid study
U_input	unquantified	several inputs assumed rather than verified
U_D	unavailable	the source reports no experimental uncertainty
U_val	does not exist	two required components are missing
Verdict	INCOMPLETE_UNCERTAINTY	

This is the single most instructive result in the repository. The computation ran, converged, and produced a clean grid study. It still cannot be compared to the experiment, because you cannot judge a difference against a band you do not have.

The error that hid here for four days. An earlier version combined only the terms that happened to be quantified and called the result U_val, giving 0.004721 and a ratio of 5.04. Dropping an unquantified component is arithmetically identical to setting it to zero, which is the one value it is known not to have. The tool now reports U_val as null when any component is missing, names the missing ones, and reports the combination of known terms under its own name so it can never be mistaken for the real thing.

7.2 A treatment sensitivity is not an experimental uncertainty

The experimental source reports results at three boundary-layer trip conditions, using 80, 120 and 180 grit. The spread across those three was used as the experimental uncertainty, 0.00441.

It is not one. Three different grits are three different experimental *conditions*, not three repeats of one condition. Their spread tells you how the result responds to changing the trip. It says nothing about how well any one condition was measured. It is not an upper bound on measurement uncertainty and it is not a lower bound either: the relationship is unknown and was never established.

Describing it as a lower bound was itself a later error, written into the report and the code on 24 September and removed on the 25th. The tool now requires the provenance of any such number to be declared alongside it, and refuses a treatment spread as experimental uncertainty however it is passed in.

7.3 A check that checked nothing

An asymptotic-range diagnostic was added, reported as 1.0066, and praised as confirmation that the grids were in the range where the convergence theory applies. An external review showed it was vacuous.

With equal refinement ratios and an order fitted from the same three values, the expression reduces algebraically to the ratio of the two finest solutions. The reported 1.0066 was exactly 1.098833 divided by 1.091606. It could not have failed for any input, because it is not a test. One set of three grids gives two *adjacent* differences, not two independent estimates.

It is retained because the value is cheap and worth seeing, but it is flagged as diagnostic, gates nothing, and a test pins the algebraic identity so nobody reads it as evidence again.

8. Comparing against other codes is not validation

The same case has published results from established solvers. Tabulating them is useful context and is not a substitute for comparison against the measurement.

Source	Lift at 10.0°	Difference from experiment
This work, finest grid	1.086740	+2.24 %
This work, Richardson extrapolated	1.088088	+2.37 %
CFL3D, Spalart–Allmaras	1.090915	+2.63 %
FUN3D, Spalart–Allmaras	1.098300	+3.33 %
CFL3D, k- ω shear stress transport	1.077800	+1.40 %
FUN3D, k- ω shear stress transport	1.084000	+1.98 %

Agreement with the reference solver to within 0.4 percent on the same grid family is real evidence that the mesh conversion, boundary conditions and solver settings are right. It is code-to-code agreement, which is verification-flavoured, not validation.

A false claim, caught by its own table. An earlier version of the report said that every published code misses this experiment by more than this work does, implying the discrepancy was the field's problem rather than the workflow's. The table above refutes it: the two shear-stress-transport references are at 1.40 and 1.98 percent, both *below* this work's 2.24. What the table does support is narrower and duller: every code listed misses in the same direction, between 1.4 and 3.3 percent, and this work sits inside that range rather than outside it.

PART III

The experiment and its statistics

How you would actually answer the question, and the several ways a reasonable-looking design quietly fails to answer it.

9. Matched thrust, and which forces you are matching

Comparing power between two shroud configurations only means something if both are doing the same job. The job is thrust, so power is compared at matched thrust rather than at matched rotational speed.

The subtlety is which thrust. A shroud is not a passive housing: the inlet lip carries suction and the duct itself produces force. Matching the *rotor's* thrust can therefore compare two assemblies producing different total useful force. The contract now specifies matched **total assembly thrust** against one declared force boundary, with rotor and duct contributions retained separately and summed.

Evidence that this is the right call. The ducted-fan benchmark audited for this project measures ducted-fan thrust and fan-rotor-only thrust as separate quantities. An experimental group working on exactly this hardware found the distinction worth instrumenting, which is stronger support for the choice than the argument above.

Electrical power and shaft power also stay distinct. Electrical power is the mean of voltage times current; shaft power is torque times angular speed. Relating them requires a stated treatment of motor and controller losses over speed, torque and temperature, and a single constant efficiency does not transfer between matched-thrust settings.

10. Reading a design matrix

Three factors, coded so that each runs from -1 to $+1$:

Code	Physical meaning	Coding
x	seam count n, one of 2, 3, 4	$x = n - 3$
y	total open angular fraction F	centred and scaled to ± 1 across its range
z	registered radial step magnitude s, $s \geq 0$	$z = 2s/s_{\text{high}} - 1$

The candidate is the thirteen unique locations of a three-factor Box–Behnken layout against a ten-column quadratic basis: a constant, three linear terms, three two-factor interactions and three squared terms. Thirteen rows for ten coefficients leaves three residual degrees of freedom.

10.1 What the audit checks, and why each check exists

Check	Result	What failure would mean
Rank	10 of 10	A coefficient could not be estimated at all
Condition number	7.109340	Small changes in data would swing the coefficients wildly
Smallest singular value	0.750093	Same, expressed as the weakest direction
Residual degrees of freedom	3	Nothing left to check the fit against

A condition number near seven is comfortable. For comparison, the design becomes numerically hopeless if two factors are nearly collinear: making one factor a near-copy of another raises it by six orders of magnitude while the rank still reports full. That is why the audit reports conditioning and not only rank, and a test constructs exactly that defect to prove it is caught.

10.2 What the design cannot see

Identifiability is not the same as completeness. The ten declared columns can be estimated; that does not make omitted effects absent. Three specific blind spots are recorded:

- **Cubic terms alias with linear ones.** At three levels coded -1 , 0 , $+1$, the cube of each factor equals the factor itself, so a pure cubic effect is indistinguishable from a linear one.
- **The three-way interaction is invisible.** The product of the three coded factors is exactly zero at every one of the thirteen training locations. A model fitted there cannot detect it at all.
- **The extreme corners are outside the hull.** Box–Behnken deliberately omits the all-extreme combinations, so no behaviour may be claimed there.

The second is handled rather than merely noted. Two of the four reserved holdouts sit where the triple product is ± 0.25 , so they are where such an effect would show. A test manufactures a surface carrying a three-way term, confirms the training fit recovers the wrong coefficients with zero residual, and confirms the holdouts expose it with exactly the predicted error.

Why zero residual is the alarming part. A model can fit its training data perfectly and be wrong about the physics. Residual error on the data you fitted is not evidence of correctness. This is the statistical form of the same mistake as code-to-code agreement standing in for validation.

10.3 Factors that are not independent

Seam count, individual seam width and total open angle are related by an exact geometric identity: total opening is count times width. Supplying all three as independent factors guarantees a singular design. Worse, varying count at fixed width also varies total open area, so an apparent count effect may be an open-area effect wearing a disguise.

The contract therefore declares two contrasts: one holding individual width fixed, one holding total opening fixed and redistributing it. The second is primary, because it isolates *arrangement* from *amount*, which is the actual research question.

11. The baseline that vanishes exactly when you need it

The natural null model predicts performance from mean clearance alone, and the defect-aware model adds seam descriptors. If the second beats the first on held-out configurations, arrangement matters.

A reviewer found that the baseline of an intercept plus mean clearance is rank-deficient precisely when the equal-mean experiment is run correctly. If every configuration has the same mean clearance by construction, that column is constant, carries no information, and the fit is singular. The design succeeding is what breaks the baseline.

The pipeline now falls back to an intercept-only baseline, which is the correct null for that design, and fits a clearance slope only where genuine clearance variation supports one.

12. Three outcomes, not two

The original pipeline could report improvement or no improvement. That is not enough, and the gap is where most over-claiming happens.

Outcome	Meaning
DEFECT_AWARE_BETTER	Beat the baseline by more than the registered gate, with acceptable prediction error
IMPROVED_BUT_ERROR_TOO_HIGH	Improved, but predictions are not accurate enough to use
RESOLVED_BELOW_GATE	A real, resolved improvement that is too small to matter
NO_IMPROVEMENT	Resolved, and in the wrong direction
PRACTICALLY_EQUIVALENT	Inside a bound declared <i>in advance</i> as "close enough"
INCONCLUSIVE	The effect lies inside its own uncertainty. Not a null result

The last two are the important additions. **Inconclusive is not a null.** An effect smaller than its own error bar is unresolved, and reporting it as evidence that arrangement does not matter is a claim the data cannot support. Practical equivalence is a real finding but only against a bound fixed before the result is seen; choosing it afterwards is circular.

The classifier is a pure function so all its branches are unit-tested, and it refuses to decide anything at all without a registered tolerance.

13. Excluding a run from a fit must not erase it

If a blade rubs the shroud, that run's steady power is meaningless and belongs out of the aerodynamic fit. But contact is an outcome this project exists to study. One field cannot carry both decisions.

Field	Values	Answers
acquisition_validity	valid · invalid · unknown	Can the measurement be trusted?
aerodynamic_eligibility	eligible · ineligible · pending	May it enter the power fit?
contact_or_clearance_outcome	no_detected_failure · failure · unknown	What physically happened?
attempted_condition_membership	included · excluded	Does it count in the denominator?

They are independent but not arbitrary. Invalid acquisition can never be eligible. A non-detection requires a detector that was working and covering the window; without that the outcome is unknown, and unknown outcomes are reported as an interval on the failure rate rather than folded in with the successes.

The denominator has to exist before the data does. The attempt register is written *before* acquisition starts. Otherwise the denominator gets rebuilt from whichever files happened to parse, and every run that failed so badly it produced no file disappears from the yield statistics. Survivor-only reporting is not a subtle bias; it is the difference between a 90 percent success rate and an unknown one.

PART IV

Evidence discipline

The part that looks like bureaucracy and is actually the method. Every rule here exists because something went wrong without it.

14. The claim ledger: what may be said out loud

One table maps each claim to its current evidence and the language that evidence permits. Some rows permit nothing.

Claim	Evidence	Language permitted now
Tip clearance affects ducted-fan performance	Primary literature	"Tip-clearance sensitivity is established"
Defect topology matters beyond mean clearance	None	"Planned comparison"
The equal-mean comparison method is itself new	Refuted by literature	Prohibited
The structure is a performance duct	None	Prohibited
The structure is a protective guard	None	Prohibited

The third row is the one to study. A close reading found a published study that holds circumferential leakage area equal to a uniform case while redistributing the gap. The method this project thought was its contribution already exists. The ledger now forbids presenting it as a contribution, and what remains distinctive has to rest on the seam geometry, the small rotor, and measurement rather than simulation.

Finding that out is a good day's work. A project that cannot record being scooped will publish a claim someone else can refute in an afternoon.

15. Provenance: where every number came from

Seven categories, applied to any number that could change a decision:

Category	Meaning
requirement	An imposed target the project must meet
measured_input	Measured, by this project or by a source of record
sourced_assumption	From an external source, <i>with its applicability condition stated</i>
calculated_result	Computed from other registered quantities
selected_value	Chosen by this project. A choice, not a derivation
measured_result	An outcome observed under a documented protocol
provisional_estimate	In force but unjustified, and marked for resolution

Evidence status is tracked on a *separate axis*: unverified, analytically assessed, physically tested, or not applicable. Keeping them apart matters because a readable full text is an access fact, not an evidence-strength fact. One literature record had been graded highly partly because its full text was available; that grade is now split, with the validated baseline and the predictions for unmeasured geometries graded separately.

The register is checked, not decorative. A script parses each registered quantity out of the source code that implements it and fails when the two disagree. It runs in continuous integration. It also refuses any quantity claiming to be physically tested, because this project has made no measurement, and it was confirmed to fail on four deliberately introduced defects rather than merely assumed to work.

15.1 Two numbers that are in force and derived nowhere

The exit gate requires a twenty percent improvement over the mean-clearance baseline and a held-out error under ten percent. Both are enforced in code. Nothing anywhere states what decision either protects or what difference is worth acting on. They are labelled provisional at every place a reader meets them, and they cannot be derived until the owner sets the minimum meaningful effect.

Finding a number like this is normal. Leaving it unlabelled is what turns a placeholder into a result.

16. Freezing acceptance before you look

Acceptance criteria for a case are written to a file and committed *before* the case runs. The file is treated as immutable and its hash is pinned by a test.

The reason is visible in the A0.1 outcome. The comparison failed against a band the project had committed to. The obvious move is to widen the band, which would have been defensible in prose and would have destroyed the result's meaning. Because the band was frozen, the only honest option was to record the failure and say why. The verdict stands and the reason is recorded.

The same logic runs through the repository. Dated progress entries are append-only history; corrections are written as new dated entries that supersede, never as edits to the old text. When a generated record could not be regenerated because its raw solver output no longer existed, the original was left byte-identical and a derived revision was added beside it, pinned to the original's hash, introducing no number that was not already there.

17. Why there are several vocabularies rather than one

The repository has five separate evidence registers: source parameters, measurement budget terms, project claims, literature access and strength, and canonical quantities. A natural instinct is to merge them.

They are deliberately separate because each governs a different kind of artifact, and merging would make every one of them vaguer. What was genuinely missing was not a sixth vocabulary but *coverage*: nothing owned a number used in more than one place, so two documents could state different values and nothing would notice. That gap is what the canonical register fills, and it fills only that gap.

PART V

The code

Seventeen scripts, nineteen test files, about 2,900 lines of source and 2,300 of tests. Standard library only, apart from two declared dependencies.

18. What each script does

18.1 Evidence and provenance

Script	Job
check_repo_contract.py	Checks the public contract: no study result is claimed, no local link is broken
check_quantities.py	Parses registered quantities out of the code and fails when register and source disagree
acquisition_ledger.py	Reconciles source routes offline. Never manufactures search provenance
reference_coverage.py	Reconciles prior-art coverage against the day-one source set
literature_search.py	Extends the corpus into themes the frozen query set never covered
literature_register.py	Builds the register from every corpus already held

18.2 Computational fluid dynamics

Script	Job
p2d_to_gmsh.py	Converts a Plot3D C-grid to a one-cell-thick mesh. Merges the wake cut and the sharp trailing edge
make_case.py	Generates one solver case. Freestream turbulence follows the published specification, not a guess
cf_d_case_manifest.py	Builds the committed record for one case: source revision, mesh hash, image digest, commit
cf_d_grid_convergence.py	Grid convergence and the validation comparison. Owns the uncertainty eligibility gate and verdict precedence
convergence_sensitivity.py	Asks whether a different convergence criterion would reclassify any recorded case

18.3 Experiment and analysis

Script	Job
analysis_pipeline.py	The whole analysis, written before any data exists. Ingestion, descriptors, matched-thrust interpolation, models, verdict
make_synthetic_fixtures.py	Generates the synthetic records the pipeline is tested against. Every file labelled not evidence
study_a_design.py	Audits the design matrix and generates its prose table from the data file
ingestion_dispositions.py	Attempt-level dispositions: what happened, kept apart from what may be analysed
transient_acceptance.py	What a steady screening result may claim once transient evidence exists
clearance_uncertainty_budget.py	The measurement budget and its fail-closed stop rule

19. Writing the analysis before the data

The analysis pipeline was written and tested against synthetic fixtures before any measurement existed. This is not merely convenient. Choosing an analysis after seeing data is how researchers talk themselves into results, and freezing it beforehand removes the opportunity.

The fixtures are built to a deliberate design: an equal-mean case, a null case, a varying-mean case, an underpowered case, one with missing data and one with a unit error. Each exercises a path the pipeline must handle correctly.

The fixtures found two real defects in the first version. The identifiability check tested each descriptor's spread but not collinearity, so total seam width, being count times a fixed width, passed and made the fit singular. And the verdict credited any positive improvement, so the null fixture reported success on a relative improvement of eight parts in ten billion.

Noiseless fixtures lie. The first fixtures had no scatter, so the uncertainty half-width came out as exactly zero and every effect looked perfectly resolved. Deterministic jitter was added. A test harness that cannot fail is the same error as the asymptotic check in Part II, in a different costume.

20. The test philosophy: prove the test can fail

A test that passes tells you nothing until you have watched it fail for the right reason. The repository uses negative controls throughout: deliberately introduce the defect, confirm the test catches it, then remove it.

This caught a bad test of my own. A check was written to confirm the code and the specification named the same verdicts, by parsing verdict assignments out of the source with an abstract syntax tree. It proved nothing about behaviour on any code path, and its document scan exempted the very file that still contained the defect. It was replaced with tests that run the real command line end to end against synthetic input.

And the controls themselves were wrong first. The first negative controls for that replacement all appeared to fail correctly. They were failing on an import error, because the tests directory is not a Python package and the runner had been invoked by dotted path. Three green-looking refutations that refuted nothing. Always read *why* a control failed, not merely that it did.

20.1 The verification gate

```
python -m unittest discover -s tests                216 tests
python scripts/check_repo_contract.py
python scripts/check_quantities.py
python scripts/study_a_design.py --check
python scripts/reference_coverage.py --check
python scripts/acquisition_ledger.py --check
python scripts/clearance_uncertainty_budget.py --check
python tools/check_presentation.py . "." ...
python tools/test_presentation.py
git diff --check
```

All of it runs in continuous integration. The contract check earned its place recently by catching a broken link in a document written minutes earlier: a date typed as the 26th where the file was dated the 25th.

PART VI

History, and every error so far

This is the curriculum. Each entry is a mistake that looked correct from the inside, which is the only kind worth cataloguing.

21. Timeline

Date	What happened
3–11 Sep	Prior-art search, database export, then a correction when the first acquisition's provenance failed audit
12–15 Sep	Reference coverage reconciled; top-25 triage; measurement requirements drafted
16 Sep	Research programme adopted: four studies, each with a kill criterion and a claim boundary
19 Sep	Validation ladder entry tasks. Source verification finds the live data site gone; specification recovered from an archive
20 Sep	A0.1 runs. Comparison against the experiment fails against the frozen band
21 Sep	Close reading narrows the novelty claim: the equal-mean method is already published
22–23 Sep	Design contract drafted; analysis pipeline written before any data
24 Sep	External review. Three defects reproduced exactly by counterexample; corrections applied
25 Sep	Owner review of the correction pull request finds it still self-contradictory. Six findings, a seventh found by the new check
26 Sep	Convergence criterion sensitivity answered. Computational restart diagnosed and blocked

22. The error catalogue

Grouped by what kind of mistake they are, because the categories transfer even when the specifics do not.

22.1 Claiming more than the evidence supports

Error	Why it looked right	What killed it
A universal two percent floor below which seam effects could not be resolved	Derived from a real, measured discrepancy in this project's own validation case	It came from absolute <i>lift</i> on a two-dimensional aerofoil and was applied to <i>differences in power</i> between ducted-rotor configurations. Different quantity, geometry and conditions; paired errors may cancel
"Every published code misses this experiment by more than this work does"	Reads as honest context-setting for a disappointing result	The report's own table: two references at 1.40 and 1.98 percent against this work's 2.24
"A0.1 validates a two-dimensional airfoil workflow"	The grid study was clean and the code-to-code agreement was good	The case's own verdict was that its uncertainty budget was incomplete
An eleven percent compressibility discrepancy predicted for the rotor case	The Prandtl–Glauert factor at that tip Mach number really is 1.11	That relation is linearised, thin-aerofoil, two-dimensional and irrotational. A rotor section has induced flow, radial flow, thickness and viscosity. It is a sensitivity flag, not a correction

22.2 Treating an unknown as a zero

Error	What killed it
Reported a validation uncertainty while a required component was unquantified	Dropping a term is arithmetically identical to setting it to zero, the one value it is known not to have
Used the spread across three grit treatments as experimental uncertainty	Three grits are three conditions, not three repeats. The spread describes the treatment, not the measurement
Then called that spread a lower bound on measurement uncertainty	It bounds nothing about the measurement in either direction. The relationship is unknown
A number passed on the command line became an uncertainty by being a number	Provenance must be declared with the value. An ineligible basis is now refused however it is supplied

22.3 Checks that could not fail

Error	What killed it
An asymptotic-range check, reported as 1.0066 and praised as confirmation	It reduces algebraically to the ratio of the two finest solutions. It was exactly that ratio
A test parsing verdict names out of source with an abstract syntax tree	It proved nothing about behaviour, and exempted the file that still held the defect
Negative controls that all failed convincingly	They were failing on an import error, not on the assertion
Noiseless synthetic fixtures	Uncertainty half-width came out exactly zero, so every effect looked perfectly resolved

22.4 Statistical and analytical mistakes

Error	What killed it
The intercept of a harmonic fit was labelled the mean clearance	On a masked domain the basis is not orthogonal to the constant. A twelve-point counterexample gave 100 against 98 micrometres
The baseline used intercept plus mean clearance	Rank-deficient exactly when the equal-mean experiment is run correctly. The design succeeding breaks the baseline
Scoring weighted rows rather than specimens	A specimen with more rows carried more weight than one with fewer
Two outcomes: improved or not	An effect inside its own uncertainty is unresolved, not absent
Seam count varied at fixed individual width	That also varies total open area, so a count effect may be an area effect
Matched rotor-only thrust	The shroud carries force itself, so two assemblies could produce different total useful force

22.5 Process and tooling

Error	What killed it
Generated results were covered by an ignore rule	Every computational record was silently untracked. Continuous integration caught it as broken links
A ledger row was dropped while resolving a merge conflict	Taking one side of a conflict wholesale. Caught by comparing against the pre-merge commit; conflicts are now resolved as a strict union by identifier
A dependency imported but never declared	A clean checkout could not rebuild the meshes
A convergence status conflating two different things	One case reached iteration 117 of a 500-iteration window. It was recorded as unconverged, which reads as a judgement nobody made

23. The one pattern behind most of them

Read the catalogue again and a single shape recurs. A quantity is computed correctly, and then used to answer a question it was never about.

Lift error answering a power-difference question. A treatment spread answering a measurement question. A fit's intercept answering a spatial-average question. Code-to-code agreement answering a validation question. Training residual answering a prediction question. In every case the arithmetic was right and the inference was not.

The habit worth stealing. Before using any number, say out loud what question it is an answer to, then say what question you are about to ask it. If those two sentences differ, stop. That single check would have caught most of the catalogue above, and it costs ten seconds.

PART VII

How to learn this, and what is still open

A study path in the order that makes each piece land, exercises against the live repository, and an honest account of what nobody has answered yet.

24. A study path

Week 1 — the distinction that organises everything

Learn verification against validation until you can state it without hesitating. Read the interpretation paper on the V&V; 20 standard first, because it makes the point this project had to learn twice: validation estimates a discrepancy for specified outputs and conditions, and is not inherently a pass-or-fail exercise. Then read the grid-convergence procedure paper and work the arithmetic in section 6 of this guide by hand.

Do not move on until you can explain why a computation that agrees with another code to within 0.4 percent is still not validated.

Week 2 — uncertainty

Read the measurement-uncertainty guide, specifically on measurement models and combining components. The idea to internalise is that an uncertainty is a statement about a measurement model, so a number without a model behind it is not an uncertainty however plausible its magnitude. That single idea explains four separate entries in the error catalogue.

Week 3 — the physics

Ducted-fan aerodynamics and tip-leakage flow. The ducted-fan benchmark audited for this project is the best single entry point, because it is experimental, it varies tip clearance deliberately, and it reports rotor-only and whole-assembly thrust separately. Note what it does *not* give: no blade section coordinates, no duct profile, no Reynolds or Mach number, and no uncertainty on the integrated coefficients. Practise reading a paper for what is absent.

Week 4 — design of experiments

Response-surface designs, aliasing and resolution. Then equivalence testing, because the difference between "we found no effect" and "we showed the effect is smaller than X" is the difference between an unsupportable claim and a publishable one.

Week 5 — read the repository in this order

- 1 docs/specs/research-programme/proposal.md — the question and the four studies
- 2 docs/specs/research-programme/plan.md — decisions D1 to D11
- 3 docs/cfd-validation-a01.md — the one case that ran, including its correction block
- 4 docs/corrections/2026-09-24-review-corrections.md — what was wrong and how it was found
- 5 docs/number-provenance-audit-2026-09-25.md — where every number came from
- 6 docs/traceability-index.md — the map from decisions to evidence

25. Exercises against the live repository

Each of these is runnable today and each teaches something the prose cannot.

#	Exercise	What it teaches
1	Reproduce the grid-convergence index by hand from the three lift values, then check against the committed record	That the procedure is arithmetic you can audit, not a black box
2	Break a registered constant in the source and run the quantity checker	What a register that is actually enforced feels like
3	Make the treatment spread eligible again and run the uncertainty behaviour tests	That the eligibility gate is real and fails for the scientific reason
4	Remove the centre point from the design file and run the design audit	How a design goes rank-deficient, and that the audit notices
5	Add a three-way interaction to the manufactured surface in the design tests	That training residual can be exactly zero while the model is wrong
6	Set a fixture to noiseless and watch the half-width collapse to zero	Why a test harness that cannot fail is worthless
7	Run the convergence sensitivity and find the case that was never assessed	The difference between failing a check and never reaching it

26. Sources worth owning

Topic	Source
Grid convergence	Celik, Ghia, Roache, Freitas, Coleman and Raad (2008), <i>Procedure for Estimation and Reporting of Uncertainty Due to Discretization in CFD Applications</i> , ASME J. Fluids Eng. 130(7). doi:10.1115/1.2960953
What validation means	Eça, Dowding and Roache (2022), <i>On the Interpretation and Scope of the V&V 20 Standard</i> . doi:10.1115/1.4053854
Uncertainty	JCGM 100, the Guide to the Expression of Uncertainty in Measurement
The A0.1 case	NASA Turbulence Modeling Resource, two-dimensional NACA 0012 validation case. The live site was gone when this project needed it; use an archive snapshot and record which
The A0.1 experiment	Ladson (1988), NASA TM 4074, on independent variation of Mach and Reynolds number for the NACA 0012
The A0.2 case	Caradonna and Tung (1981), NASA TM 81232, hovering rotor experiment
Ducted fans and tip clearance	Akturk and Camci, GT2011-46356, ASME Turbo Expo 2011; journal version J. Turbomach. 136(2):021004 (2014)
Non-uniform clearance	Graf, Wong, Greitzer, Marble, Tan and Shin, J. Turbomach., issue date October 1998. doi:10.1115/1.2841774. Unread here: a green open-access copy is indexed at MIT but sits behind a verification page
Response-surface designs	NIST/SEMATECH e-Handbook of Statistical Methods, Box–Behnken designs and the sections on resolution and aliasing
Equivalence testing	Lakens (2017), a practical primer on equivalence tests

27. What is open, and who owns it

27.1 Waiting on you

Item	What it unblocks
The minimum meaningful power difference, and the clearance-setting requirement	The two acceptance gates, which are currently enforced and derived nowhere
Target mean clearance and the rotor, stand and sensor envelope	Physical factor levels, the measurement budget, and every CAD task
The solver regime for the rotor case	The second validation rung. A verified compressible workflow is recommended for any pressure claim, and that recommendation is not adopted
The public disclosure boundary	Two withheld tasks, and CAD generally
Institutional access, or a browser session	The non-uniform clearance paper, which is the highest-value unread source

27.2 Agent work, not blocked

- Separating a terminated run from an assessed-and-unconverged one in the status vocabulary
- A stated basis for the identifiability tolerances
- The remaining unselected literature records, the under-sampled themes, and a dated patent search

27.3 Blocked by the physical world

The restart of the unfinished turbulence-model arm is diagnosed and blocked. No checkpoint, mesh or raw output survives, so a restart would be a fresh run whose regenerated mesh must hash to the recorded value or the grid family breaks. The host is also under worse memory pressure than when the run first stalled.

All ten computer-aided design tasks are deferred and every one is transitively gated. Building any of them would require inventing a rotor radius, a mean clearance and a seam geometry that no decision has set.

28. The honest position, in one paragraph

This project has a well-posed question, a defensible method, a complete analysis written before any data, an audited design matrix, and a validation case that ran cleanly and still cannot be compared to its experiment because the experiment reports no uncertainty. It has no measurement, no built hardware, and a novelty claim that is narrower than it started. Its most valuable output so far is the discipline that caught roughly twenty of its own errors, several of which had already been written down confidently, published in a pull request, and passed continuous integration before anyone noticed. A green check mark is not a finding.

Guide generated 27 September 2026 from the repository at commit 2317d8f. Every number quoted here was read from the committed record or recomputed from it. Nothing in this document is a measurement.